Class-Aware Consistency Learning for Open-Set Semi-Supervised Hyperspectral Image Classification

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***Abstract*—Semi-supervised hyperspectral image (HSI) classification methods focus on exploring the spectral and spatial information of unlabeled samples. However, existing methods generally follow the closed-set setting, assuming that unlabeled samples do not contain novel classes, which is hard to hold in practical applications. This article aims to study semi-supervised HSI classification in the open-set setting, i.e., unlabeled samples fall into novel classes, and proposes a class-aware consistency learning (CACL) method. First, to explore discriminative spectral–spatial features, a position-aware transformer (PAT) is developed, which effectively models spatial position priors between the center pixel and its neighboring pixels via a symmetric position-aware encoding (PAE). Then, to reduce the interference from novel class samples on the model's discrimination, a prototype-driven consistency learning (PDCL) is proposed, which accurately selects unlabeled samples belonging to known classes via a known class sampler, and efficiently utilizes their spectral–spatial information by modeling consistent predictions across different views. Finally, to further improve the distinguishability between known classes, a prototype contrastive optimization (PCO) is proposed to decrease the distance between samples from the same class and increase the distances between those from different classes in the feature domain. Furthermore, an adaptive segmentation threshold is designed to accurately predict known classes and reject novel classes. Extensive experiments verify that our CACL outperforms the state-of-the-art methods, achieving the overall accuracy (OA) of 81.65%, 88.61%, and 92.88% on the Indian Pines (IP), Salinas (SA), and Pavia University (PU) datasets, with ten labeled samples in each known class. The code is available at https://github.com/rock-in/CACL-main.**

***Index Terms*—Class prototype, hyperspectral image (HSI) classification, open-set classification, semi-supervised learning (SSL).**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain rich spectral and spatial information of observed targets [1], [2], [3], [4]. These spectral signals and spatial information can be used to identify the land-cover classes of observed targets [5], [6], [7]. HSI classification aims to identify the class of each pixel by analyzing its spectral–spatial information, which plays a key role in environmental monitoring, precision agriculture, natural resource management, and urban planning [8], [9], [10], [11]. With the rapid development of deep learning, numerous methods employ various neural networks (e.g., convolutional neural networks (CNNs), vision transformer (ViT), and Mamba) to extract spectral–spatial features for analyzing the classes of pixels [12], [13], [14]. These methods achieve promising classification results when there are sufficient labeled samples. However, the classification performance drops sharply under limited labeled samples [15], [16], [17], [18].

Semi-supervised learning (SSL) HSI classification methods leverage both labeled and unlabeled samples to train models, reducing reliance on labeled samples and achieving admirable performance [19], [20]. These methods can be roughly divided into graph learning [20], [21], contrastive learning [22], [23], reconstruction [24], [25], and pseudo-label learning methods [19], [26]. Graph learning methods treat pixels or small regions as graph nodes to construct topology structures for the HSI, and use graph neural networks (GNNs) to extract features [21]. Contrastive learning methods design pretext tasks to learn spectral–spatial information by making similar samples closer and pushing dissimilar samples far from each other [23]. Reconstruction methods mine potential information of unlabeled samples by minimizing the spectral–spatial differences between inputs and reconstructed outputs, and then exploit a few labeled samples to optimize the model's parameters [24]. Pseudo-label learning methods utilize labeled samples and unlabeled samples with confidence scores higher than a threshold to train classification models [19].

Existing semi-supervised HSI classification methods imply a prior assumption that unlabeled samples in both training and testing sets share the same land-cover classes as labeled samples, as shown in Fig. 1. In other words, they assume unlabeled samples do not contain novel classes that have not been seen in labeled samples. These methods are named as *closed-set semi-supervised HSI classification* in this article [9],

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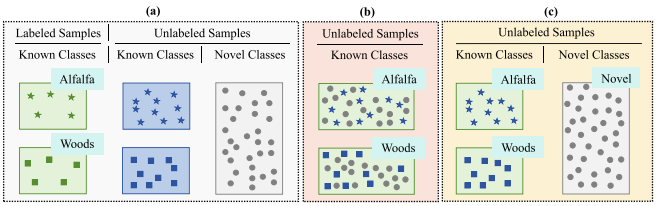


Fig. 1. Comparison of closed-set and open-set semi-supervised HSI classification. (a) Unlabeled samples contain known classes and novel classes. (b) Closed-set semi-supervised classification classifies unlabeled samples belonging to novel classes as known classes in the training and testing stages. (c) Open-set semi-supervised classification uses both labeled and unlabeled samples to train a classification model, which is subsequently used to predict known classes and reject novel ones.

[19], [27], [28]. However, due to the diversity of land-cover classes, limited classes in labeled samples cannot fully cover the ground truth of all unlabeled samples, resulting in these unlabeled samples often containing novel classes. Therefore, existing closed-set semi-supervised HSI classification methods are hard to hold in practice. To solve this issue, this article introduces *open-set semi-supervised HSI classification*, which aims to use unlabeled samples containing novel classes to train a classification model, and this model is expected to accurately predict known classes and reject novel classes [29], [30].

To accurately identify the land-cover classes of unlabeled pixels, two main challenges need to be addressed in open-set semi-supervised HSI classification. One is "How to effectively leverage unlabeled samples in the training stage?" Due to the diversity of land-cover classes, hyperspectral pixels within the HSI have complex spatial position distribution information, especially those located in edge areas. Furthermore, affected by the imaging background (e.g., changes in light, temperature, and angle), unlabeled pixels from novel classes may also have similar spectral bands with those from known classes [31], [32]. If unlabeled samples belonging to novel classes are used to train the model's parameters, these samples may confuse the model's discrimination when labeled samples are small, thereby reducing the classification results. Another is "How to accurately identify the classes of unlabeled samples in the testing stage?" In real scenarios, unlabeled samples in the testing set also contain novel classes. Accurately identifying the land-cover classes of these unlabeled samples is challenging.

To address these challenges, a class-aware consistency learning (CACL) method is proposed, which effectively utilizes the spectral–spatial information of known class samples for open-set semi-supervised HSI classification. First, traditional transformer-based methods model the spatial position information of input tokens via a learnable positional encoding (PE). These methods often overlook spatial position priors between the center pixel and its neighboring pixels, resulting in insufficient utilization of spectral–spatial information. To extract rich and discriminative features, a position-aware transformer (PAT) is developed, which effectively models spatial position priors between pixels via a symmetric position-aware encoding (PAE). Then, to reduce novel class samples interfering with the distinguishability of known classes, a prototype-driven consistency learning (PDCL) method is introduced, which accurately selects unlabeled samples belonging to known classes via a known class sampler, and exploits their spectral–spatial information by modeling consistent predictions. Finally, to further improve the distinguishability between known classes, a prototype contrastive optimization (PCO) method is introduced, which pulls samples from the same class together and pushes away those from different classes. Moreover, to alleviate the tediousness of hand-designed thresholds, an adaptive segmentation threshold is proposed to predict known classes and reject novel ones in the testing stage. The main contributions of this article are concluded as follows.

1) This article studies the performance of existing semi-supervised HSI classification methods in open-set settings and proposes a CACL method to efficiently exploit unlabeled samples to improve classification results.
2) To eliminate the interference from novel class samples on the model's discrimination, a PDCL is introduced to selectively utilize the spectral–spatial information of unlabeled samples via a known class sampler.
3) To further improve the distinguishability between known classes, a PCO is proposed to constrain samples from the same class closer and push those from different classes away. Furthermore, an adaptive segmentation threshold is designed to accurately identify known classes and reject novel classes.

The rest of this article is organized as follows. Section II recalls related works. Section III describes our proposed CACL in detail. Section IV introduces experimental settings and results. Finally, Section V concludes this article.

II. Related Work

A. Closed-Set Semi-Supervised HSI Classification

Due to powerful representation capabilities, numerous methods use various neural networks to extract features from labeled samples [16], [17], [32]. These methods extract discriminative features when numerous labeled samples are available. However, the accuracy drops sharply when labeled samples are scarce. SSL methods further improve the performance and the model's generalization by utilizing the spectral–spatial information of unlabeled samples [19], [20], [33], [34], which are mainly categorized into four groups: graph learning methods, contrastive learning methods, reconstruction methods, and pseudo-label learning methods.

Graph learning methods often construct a topology structure for the HSI, and then extract features using GNNs [20], [21], [35], [36]. According to the constructed manners of topology structure, these methods can be divided into pixel and superpixel methods. Pixel methods treat pixels as graph nodes to construct an adjacency matrix, and then extract features for identifying the pixel's classes. Since the spatial resolution of the HSI is large, pixel methods often require a lot of computing resources and are easily affected by noise perturbations. Superpixel methods generally split the HSI into a series of irregular regions, and treat these irregular regions as graph nodes. Xi et al. [21] designed a multigraph to represent the topology of the HSI. Zhou et al. [35] extracted global and local features using GNNs and CNNs. Lu et al. [20]

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introduced a multiscale superpixel segmentation strategy to generate pseudo-labels, and combined consistency learning to leverage their spectral–spatial information. Hu et al. [36] transformed the HSI into an edge-masked perturbed graph to exploit the multiscale features without supervised information. Although superpixel methods exhibit noise immunity, their classification results are affected by the quality of graph structures.

Contrastive learning methods extract the spectral–spatial information of the HSI by designing pretext tasks, which aim to make similar samples as close as possible and push dissimilar samples far away [22], [23], [37], [38], [39]. Yu et al. [22] combined GNNs with contrastive learning to extract features. Cai et al. [37] used the transformer to extract features of the HSI and LiDAR, and then optimized the model's parameters using a contrastive loss computed from different modalities. Xu et al. [38] extracted features by aligning projected features of different views belonging to the same unlabeled sample. Although these methods effectively utilize the spectral–spatial information of unlabeled data, inappropriate data augmentation methods and negative sample selection strategies will decrease classification accuracy.

Reconstruction methods utilize the spectral–spatial information of unlabeled samples by minimizing the spectral–spatial differences between inputs and reconstructed outputs, and then they optimize the model's parameters by using a few labeled samples [24], [25], [40]. Jia et al. [24] employed the reconstruction loss of unlabeled samples to pretrain a classification model consisting of 3-D CNNs, and then fine-tuned the model's parameters using a small number of labeled samples. Cai et al. [41] proposed a hypergraph-structured autoencoder to utilize high-order relations between samples, which can extract robust features for classification. Although the model's parameters are further fine-tuned with labeled samples, these methods focus on reconstructing inputs, which results in existing methods struggling to capture discriminative features for classification under a few labeled samples.

Pseudo-label learning methods generally preserve unlabeled samples with confidence scores larger than a preset threshold, and treat the maximum predictions of confidence scores as pseudo-labels [26], [34], [42], [43], [44], [45], [46]. Wu and Prasad [26] leveraged unlabeled samples with confidence scores higher than a threshold to train models, and then fine-tuned the model's parameters through a few labeled samples. Yao et al. [19] designed a novel contrastive learning paradigm to exploit low-confidence unlabeled samples, which are treated as candidate negative samples for corresponding classes. Chen et al. [9] designed adaptive thresholds for different classes to generate high-quality pseudo-labels, and used class prototypes constructed by labeled and unlabeled samples to filter incorrect labels. Xu et al. [34] designed a self-ensembling framework consisting of an ensemble network and a base network, which utilizes pseudo-labels generated by the ensemble network to optimize the weight parameters of the base network.

Existing semi-supervised HSI classification methods achieve admirable classification accuracy by leveraging the spectral and spatial information of massive unlabeled sample data. However, these methods are generally performed in a closed environment, and implicitly assume that unlabeled samples in both training and testing sets do not contain novel classes. Due to the continuous changes of the land-cover classes, unlabeled samples may fall into novel classes, which results in existing closed-set semi-supervised HSI classification methods being difficult to hold, and their classification accuracy dropping sharply in open scenarios.

B. Open-Set HSI Classification

To accurately identify known classes and reject novel classes, numerous image classification methods have been proposed [30], [47], [48], [49], [50], [51], [52]. Yoshihashi et al. [48] proposed to optimize the model's parameters by using the reconstruction and classification of training samples, and utilize the reconstruction loss of unlabeled data to distinguish known and novel classes. Wang et al. [49] proposed to learn diverse representations by constraining attention maps learned by different experts, and designed an attention diversity regularization loss to improve the richness of these experts. To utilize unlabeled sample data, Chen et al. [50] proposed an uncertainty-aware self-distillation method, which discards novel class samples based on the soft labels generated by ensemble prediction. Saito et al. [51] proposed to identify known and novel classes through one-vs-all classifiers, and introduced an open-set soft-consistency regularization loss to enhance the smoothness of these classifiers. Although these methods achieve promising classification accuracy, they are primarily designed for RGB images, neglecting the complex spectral and spatial information in the HSI. Therefore, they fail to directly apply to the HSI classification.

Several HSI classification methods performed in the open environment have been proposed, which are split into reconstruction loss and prototype learning methods [53], [54], [55], [56], [57], [58], [59]. Reconstruction loss methods assume that the model can accurately reconstruct the spectral–spatial information of samples from known classes, and fail to reconstruct those belonging to novel classes. These methods generally utilize labeled samples to train an encoder–decoder network, which is further used to reject novel classes by analyzing the reconstruction loss between inputs and reconstructed outputs in the inference stage. Liu et al. [53] designed an encoder–decoder to extract spectral–spatial features in a reconstruction manner, and then combine extreme value theory to reject novel classes based on the reconstruction losses of unlabeled samples. Yue et al. [54] designed a dual-branch decoder to reconstruct spectral and spatial information, respectively. Samples from the same class cluster together in the feature space, and those from different classes are separated distinctly [9], [56]. To this end, prototype learning methods construct class prototypes for each known class using labeled sample features, and then predict known classes and reject novel classes based on the deviation level between unlabeled sample features and class prototypes. Sun et al. [55] designed a spectral–spatial multilayer perceptron (MLP)-like as a backbone to extract features of the HSI, and

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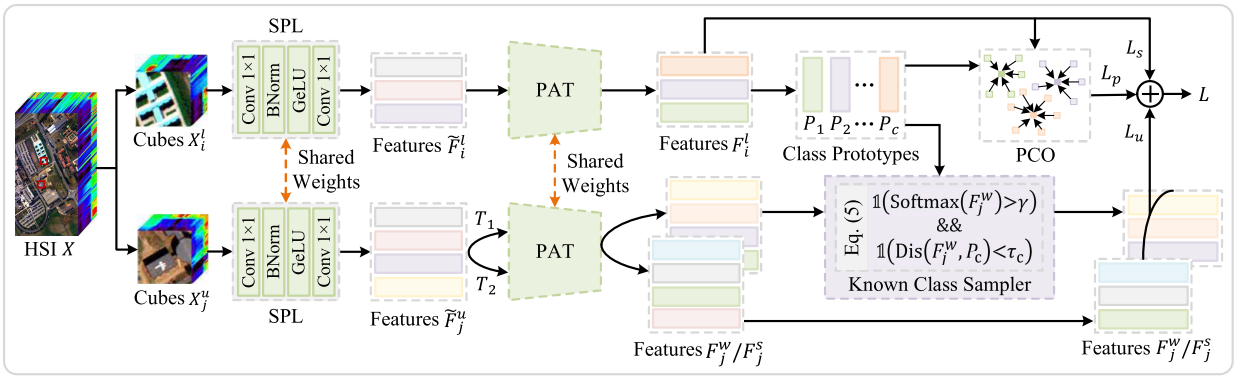


Fig. 2. Detailed framework of the proposed CACL.

introduced reciprocal point learning for open-set classification. Xie et al. [56] constructed a fixed-length class prototype set using labeled samples for different known classes, and then utilized the cosine similarity between known class prototypes and unlabeled sample features to predict known classes and reject novel classes.

Existing open-set HSI classification methods effectively extract the spectral–spatial features of the HSI, and improve classification accuracy when numerous labeled samples are available. However, when the number of labeled samples is small, these methods are prone to overfitting and fail to effectively learn spectral and spatial differences between known and novel classes. In addition, these methods only utilize labeled samples to train the model, and ignore diverse spectral–spatial information contained in unlabeled samples, which results in the underutilization of spectral–spatial information. To solve these problems, a CACL method is proposed, which effectively exploits the spectral–spatial information of unlabeled samples to optimize the model's parameters, improving the feature discrimination and the classification accuracy.

III. Proposed Method

A. Overview

In this article, the HSI is represented by $\mathcal{X} \in \mathbb{R}^{H\times W\times D}$, where H, W, and D denote the height, the width, and the number of spectral bands, respectively. The proposed CACL's weight parameters are trained by using both labeled samples $\mathcal{D}_l = \{(x_i^l, y_i^l)\}_{i=1}^{|\mathcal{D}_l|}$ and unlabeled samples $\mathcal{D}_u = \{(x_j^u)\}_{j=1}^{|\mathcal{D}_u|}$, where $x_i^l/x_j^u \in \mathbb{R}^D$ represent the ith/jth labeled/unlabeled pixels, $y_i^l \in \mathbb{R}^C$ denotes the land-cover class of x_i^l, C indicates the number of known classes. As neighboring pixels can provide rich spectral–spatial information for the center pixel, we crop HSI cubes $X_i^l/X_j^u \in \mathbb{R}^{q\times q\times D}$ centered on the ith/jth labeled/unlabeled pixels as inputs, where q represents the spatial size of HSI cubes.

To accurately identify known classes and reject novel ones, this article proposes a CACL method that selectively utilizes the spectral–spatial information of unlabeled data for model optimization. The detailed framework of the proposed CACL is illustrated in Fig. 2. First, as adjacent spectral bands have significant correlations, a spectral projection layer (SPL) consisting of two 1×1 convolutional blocks with the GeLU activation function and the batch normalization is applied to eliminate spectral redundancy, thereby producing shallow spectral–spatial features. Then, to effectively explore discriminative spectral–spatial features, a PAT is developed to model spatial position priors between the center pixel and its neighboring pixels via a symmetric PAE. Third, to further improve the feature discrimination, a PDCL is introduced, which accurately selects unlabeled samples belonging to known classes via a known class sampler, and then efficiently leverages the spectral–spatial information of these retained samples by constraining the predictions to be consistent across different views. Furthermore, a PCO is introduced to improve interclass distinguishability and intra-class compactness by restricting samples from different classes far from each other and decreasing the distance between samples from the same class in the feature domain.

B. Position-Aware Transformer

Previous works [9], [19] demonstrate that the transformer can effectively model long-range spectral–spatial dependencies of input tokens through the attention mechanism. For labeled samples X_i^l, the spectral–spatial features extracted by the transformer are described as

$$\begin{cases} F_i^{l,0} = \text{Concat}\left[\text{cls}; f_1^l, f_2^l, \ldots, f_{q^2}^l\right] + E^{\text{PE}} \\ z_i^{l,m} = \text{LN}\left(\text{MHSA}\left(F_i^{l,m-1}\right) + F_i^{l,m-1}\right) \\ F_i^{l,m} = \text{LN}\left(\text{FFN}\left(z_i^{l,m}\right) + z_i^{l,m}\right) \end{cases} \tag{1}$$

where Concat[·] represents the concatenation operation, $f_i^l \in \mathbb{R}^D$ denotes the pixel token within the shallow spectral–spatial features $\tilde{F}_i^l$ extracted by the SPL, cls $\in \mathbb{R}^D$ represents the learnable class token, $E^{\text{PE}} \in \mathbb{R}^{(q^2+1)\times D}$ indicates the learnable PE, LN(·) represents the layer normalization, MHSA(·) denotes the multihead attention mechanism, and FFN(·) represents the feed-forward neural network. For labeled samples X_i^l, the cross-entropy loss function is used to calculate the supervised loss, which is defined as

$$L_s = \frac{1}{B}\sum_{i=1}^{B} \text{CroEnt}\left(p_i^l, y_i^l\right) \tag{2}$$

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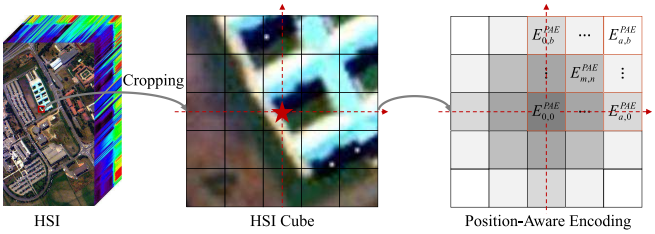


Fig. 3. Illustration of the proposed PAE.

where B indicates the batch size, CroEnt$(\cdot)$ denotes the cross-entropy loss function, and $p_i^l = \text{Softmax}(F_i^l) \in \mathbb{R}^C$ represents the confidence score of X_i^l generated from a softmax classifier.

As neighboring pixels within HSI cubes X_i^l/X_j^u often belong to the same class, these pixels can provide rich spectral and spatial information for their center pixels. Existing transformer-based methods model the spatial position information of input tokens via a learnable position encoding. However, the learnable position encoding cannot effectively exploit spatial position priors between the center pixel and its neighboring pixels, thereby limiting the feature discrimination. On the one hand, neighboring pixels closer to the center pixel are more likely to exhibit the same class, and they are more crucial for predicting the class of the center pixel. On the other hand, due to the symmetry of neighboring pixels in the spatial domain, neighboring pixels at similar positions should have the same spatial position information.

To solve this problem, a PAT is built to extract robust and discriminative features by modeling spatial position priors between different pixels within an HSI cube via the proposed symmetric PAE. As shown in Fig. 3, we take the orange area (first quadrant) as an example to introduce the proposed PAE. The proposed CACL uses a learnable encoding E_h to model the spatial position relations between the center pixel and its neighboring pixels in the horizontal direction (called horizontal neighboring pixels). The spatial position relations between the center pixel and its neighboring pixels in the vertical direction (named vertical neighboring pixels) can also be represented by E_v. Due to neighboring pixels generally belonging to the same category as the center pixel, these neighboring pixels located at similar positions have the same spatial position information, indicating that their PEs should also be the same, i.e., $E_h = E_v$. For the remaining neighboring pixels at the position (m, n), the spatial position information is generated from the horizontal neighboring pixels $(m, 0)$ and the vertical neighboring pixels $(0, n)$. The proposed PAE is described as

$$\begin{cases} E_h = \left[E_{0,0}^{\text{PAE}}, \ldots, E_{m,0}^{\text{PAE}}, \ldots, E_{a,0}^{\text{PAE}}\right] \\ E_v = \left[E_{0,0}^{\text{PAE}}, \ldots, E_{0,n}^{\text{PAE}}, \ldots, E_{0,b}^{\text{PAE}}\right] \\ E_{m,n}^{\text{PAE}} = E_{m,0}^{\text{PAE}} + E_{0,n}^{\text{PAE}} \end{cases} \tag{3}$$

where $a/b = \lfloor q/2 \rfloor$, q represents the spatial size of HSI cubes, and $\lfloor \cdot \rfloor$ indicates the floor function.

Since the spatial position relations between the neighboring pixels and its center pixel in other quadrants are similar to those in the first quadrant, the spatial position encoding of neighboring pixels and the center pixel in these quadrants are the same as those in the first quadrant. In this way, we can ensure the symmetry of the proposed PAE while effectively modeling the spatial position priors of the HSI.

C. Prototype-Driven Consistency Learning

Due to the continuous change of land-cover classes and the high cost of manual labeling, unlabeled samples may fall into novel classes in real-world applications. Although those from novel classes also contain abundant spectral and spatial information, if these novel class samples are used to optimize the model's weight parameters, this model may be confused, and the learned spectral–spatial features struggle to be distinguished. On the one hand, affected by imaging conditions such as temperature, illumination, and humidity, the HSI exhibits complex spectral information. Hyperspectral pixels belonging to different classes have similar spectral bands, and those from the same class exhibit differences [31], [32], [60]. On the other hand, due to diverse land-cover classes, hyperspectral pixels show complex spatial position distribution information, especially those located in edge regions. When labeled samples are scarce, it is challenging to learn the subtle differences between samples in spatial position distribution and spectral bands, which reduces the model's discrimination and interferes with the distinguishability between known classes, thereby failing to identify known classes and reject novel ones.

To address this problem, a PDCL is proposed to selectively exploit the spectral–spatial information of unlabeled samples, in which a known class sampler is designed to accurately filter unlabeled samples belonging to novel classes, and a class-aware consistency loss is then introduced to efficiently exploit the spectral–spatial information of the remained unlabeled samples by constraining the predictions from different views (weak augmentation T_1 and strong augmentation T_2) to be consistent. In this way, the proposed method can effectively utilize unlabeled samples from known classes to optimize the model's weight parameters, while reducing interference from novel class samples. The proposed class-aware consistency loss is described as

$$L_u = \frac{1}{|D_u^k|} \sum_{j=1}^{\mu B} \text{CroEnt}\left(p_j^w, p_j^s\right) \cdot \mathcal{F}\left(X_j^u\right) \tag{4}$$

where μ represents the relative ratio between unlabeled and labeled samples, CroEnt$(\cdot)$ indicates the cross-entropy loss function, $\mathcal{F}(\cdot)$ denotes the proposed known class sampler, D_u^k represents the unlabeled sample set belonging to known classes selected by the proposed known class sampler, and $p_j^w/p_j^s \in \mathbb{R}^C$ indicate the confidence scores of weak and strong augmentation views, respectively.

To filter unlabeled samples belonging to novel classes, a common practice is to regard unlabeled samples with confidence scores lower than a confidence threshold as novel classes, and the remaining unlabeled samples are treated as known classes. However, due to the complex spectral–spatial information, existing methods are prone to overfitting when labeled samples are scarce. Unlabeled samples belonging to novel classes may also have high confidence scores in known

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classes, which results in the misclassification of novel classes as known classes. Previous studies have shown that samples from the same class are closer in the feature space, and those from different classes are dispersed [9]. To this end, the known class sampler is introduced to integrate the confidence score with the deviation level between unlabeled sample features and class prototypes to select unlabeled samples belonging to known classes accurately, which is described as

$$\mathcal{F}\left(X_j^u\right) = \mathbb{1}\left(p_j^w > \gamma\right) \cdot \mathbb{1}\left(\mathrm{Dis}\left(F_j^w, P_c\right) < \tau_c\right) \tag{5}$$

where $\mathbb{1}(\cdot)$ represents the indicator function, γ indicates the confidence threshold, τ_c denotes the deviation threshold of the cth class, $\mathrm{Dis}(\cdot)$ is the Euclidean distance function, $P = \{P_1, \ldots, P_c, \ldots, P_C\} \in \mathbb{R}^{C\times D}$ are the known class prototypes, P_c is the cth class prototype calculated from labeled sample features using the channel-wise average operation, and p_j^w and F_j^w are the confidence score and spectral–spatial features of weak augmentation view, respectively.

Due to the complex data distribution of the HSI, it is unreasonable to set the same deviation threshold τ_c for different classes. To accurately filter novel class samples, an adaptive deviation threshold is introduced, which calculates the deviation threshold τ_c of the cth class by modeling the distance between sample features and their corresponding class prototype as a Gaussian distribution [61]. The mean μ_c and standard deviation σ_c are calculated based on the distances between labeled sample features of the cth class and the corresponding class prototype P_c. The deviation threshold τ_c of the cth class is described as

$$\tau_c = \mu_c + \beta\sigma_c \tag{6}$$

where β represents the bias coefficient and is set to 2.

In addition, the HSI often has complex spatial information as the diversity of land-cover classes. Moreover, due to the continuous changes of imaging conditions, spectral bands of hyperspectral pixels exhibit intra-class difference and inter-class similarity. The complex spectral–spatial information will easily cause classification models to confuse known classes with novel classes. To improve interclass distinguishability and intra-class compactness, a PCO is introduced to attract samples from the same class closer and push those from different classes further away, which is described as

$$L_p = \frac{1}{\mathrm{CB}} \sum_{c=1}^{C} \sum_{i=1}^{B} \log \frac{e^{-\mathrm{Dis}\left(F_i^l, P_c\right)^2/T}}{\sum_{k=1}^{C} e^{-\mathrm{Dis}\left(F_i^l, P_k\right)^2/T}} \tag{7}$$

where T denotes the temperature coefficient and is set to 0.1.

In this article, the optimization goal of our proposed CACL is to minimize the loss function, which is described as

$$L = L_s + \lambda_u \cdot L_u + \lambda_p \cdot L_p \tag{8}$$

where λ_u and λ_p represent the tradeoff coefficient.

D. Open-Set Semi-Supervised HSI Classification

Previous studies have shown that samples from the same class are compact in the feature domain, and those from different classes are scattered. Based on this phenomenon, our proposed CACL also employs the deviation level between unlabeled sample features and known class prototypes to distinguish known classes and reject novel classes. Specifically, if an unlabeled sample deviates from all known class prototypes, this sample is labeled as novel classes, otherwise as known classes. The novel class mask is described as

$$m_j^u = \begin{cases} 1, & \mathrm{Dis}\left(F_j^u, P_c\right) > \theta_c \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $m_j^u = 1$ means that the jth unlabeled sample is classified as novel classes, F_j^u is the spectral–spatial features of X_j^u, and θ_c is the segmentation threshold of the cth class. Therefore, the predicted label $\hat{y}_j^u$ of X_j^u is described as

$$\hat{y}_j^u = \begin{cases} \mathrm{argmax}\left(p_j^u\right), & m_j^u = 0 \\ \text{novel}, & m_j^u = 1. \end{cases} \tag{10}$$

Due to the complex spatial position distribution and spectral information of the HSI, setting an appropriate segmentation threshold θ_c based on experience is challenging. To this end, an adaptive segmentation threshold θ_c for different known classes is proposed to reject novel classes. Specifically, for the cth class, our proposed CACL first fits the deviation levels via a Gaussian distribution $\mathcal{N}(\mu_c, \sigma_c)$ [61], which is computed from the labeled sample features and the corresponding class prototype through the Euclidean distance function. Then, CACL treats the remaining labeled samples from other classes as negative samples of the cth class, and also uses another Gaussian distribution $\mathcal{N}^-(\mu_{c^-}, \sigma_{c^-})$ to fits the deviation levels of these negative samples. Finally, the value of θ_c corresponding to the intersection between $\mathcal{N}(\mu_c, \sigma_c)$ and $\mathcal{N}^-(\mu_{c^-}, \sigma_{c^-})$ is treated as the segmentation threshold of the cth class.

IV. Experiment

A. Compared Method and Dataset

To demonstrate the effectiveness of our proposed CACL, nine state-of-the-art methods are employed as compared methods, including SpectralFormer [16], SSFTT [13], MDL4OW [53], SSMLP-RPL [55], FullyContNet [33], PUSL [19], AMGCFN [35], RSEN [34], and SGMAEs [36]. The experimental setups of MDL4OW and SSMLP-RPL follow their original papers. For the remaining methods, this article regards confidence scores of unlabeled samples larger than 0.5 as known classes, otherwise as novel classes. To evaluate the CACL's classification accuracy, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient in the open environment are used as evaluation metrics. In addition, we conduct extensive experiments on the Indian Pines (IP), Salinas (SA), and Pavia University (PU) datasets.

B. Experimental Setting

The feature dimension of the proposed CACL is set to 64, and the number of heads in MHSA is set to 4. The relative ratio μ between unlabeled and labeled samples is set to 4, and the batch size is set to 128. The number of iterations is set to 10 000 on the IP, SA, and PU datasets. The Adam optimizer with $\epsilon = 1e^{-8}$, $\beta_1 = 0.9$, and $\beta_2 = 0.999$ is employed, and a fixed learning rate of 0.0005 is applied to update the model's

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TABLE I
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE IP DATASET

Size	Class	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	FullyContNet	PUSL	AMGCFN	RSEN	SGMAEs	CACL
10	1	79.57	96.52	91.74	95.00	98.91	96.52	**99.57**	86.96	98.04	95.56
	2	32.04	51.62	52.21	66.30	60.41	70.70	70.29	73.17	40.36	**77.63**
	4	54.68	78.82	80.34	86.74	95.91	86.33	**98.73**	49.54	80.38	87.00
	5	52.46	77.14	79.42	83.74	78.12	79.09	83.21	67.33	86.60	**89.11**
	7	97.14	**100.00**	98.93	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	98.89
	8	88.58	92.30	85.52	95.62	98.74	95.69	99.67	**100.00**	93.26	97.52
	9	99.00	**100.00**	97.50	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
	10	46.85	62.94	67.05	74.31	81.02	76.48	73.90	34.03	63.85	**83.35**
	11	53.15	66.24	56.06	67.87	72.96	75.88	76.66	67.34	57.60	**80.36**
	12	36.26	45.16	65.48	70.48	68.55	64.05	80.69	53.46	62.92	**86.31**
	13	92.88	99.12	96.63	98.10	**99.71**	98.44	98.34	96.68	98.68	95.69
	14	55.64	86.74	85.17	86.62	**92.43**	86.96	91.04	79.15	89.74	89.27
	15	33.47	81.97	78.16	87.85	**94.72**	94.46	91.11	81.94	88.55	88.43
	16	95.27	99.35	86.34	99.16	99.57	99.35	98.17	96.88	**100.00**	99.04
	Novel	11.09±1.82	6.19±1.13	0.29±0.28	9.71±4.29	0.03±0.08	4.35±3.24	38.97±10.35	0.48±0.66	15.39±3.67	**65.67±3.23**
	OA	44.96±3.34	59.93±1.36	57.53±5.08	66.01±2.67	66.84±1.59	67.90±1.01	75.06±4.22	58.23±2.58	59.95±1.27	**81.65±3.48**
	AA	61.87±2.32	76.28±1.39	74.72±4.31	81.43±1.52	82.74±0.30	81.89±1.53	86.69±2.21	72.46±0.93	78.36±0.53	**88.92±1.79**
	Kappa	37.91±3.45	54.33±8.11	52.24±5.21	65.11±2.63	62.69±1.73	63.62±1.18	71.50±4.68	52.24±2.65	54.81±1.27	**78.78±4.00**
20	1	91.74	97.10	98.04	99.23	99.78	**100.00**	99.78	86.74	**100.00**	99.62
	2	47.62	55.59	74.10	76.54	77.59	81.02	86.45	80.20	71.59	**88.15**
	4	79.92	84.81	95.49	96.50	98.35	96.96	**99.66**	79.83	91.82	98.02
	5	72.05	84.75	88.88	89.09	90.70	86.79	88.92	87.02	92.03	**92.51**
	7	98.57	99.40	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.44
	8	95.06	94.35	97.72	96.88	99.98	99.79	99.92	**100.00**	97.62	99.52
	9	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
	10	72.24	76.85	78.85	83.61	88.62	85.78	88.64	85.07	77.25	**92.00**
	11	52.15	56.30	64.60	76.61	81.43	77.96	**89.21**	77.83	72.46	87.24
	12	50.39	58.26	87.84	85.62	90.08	85.46	92.34	61.26	68.32	**92.86**
	13	98.54	98.21	98.29	99.03	99.61	**99.80**	99.61	96.63	99.51	98.59
	14	70.64	87.42	86.30	90.51	94.66	93.50	95.75	80.61	91.26	**97.07**
	15	61.55	88.17	93.03	93.06	98.24	96.63	**99.17**	90.31	96.02	95.93
	16	97.20	98.03	85.27	99.32	99.35	99.57	99.35	98.28	**100.00**	99.18
	Novel	8.38±1.93	3.35±1.60	1.03±0.29	21.51±6.55	0.03±0.05	2.81±1.24	27.99±9.10	0.28±0.25	6.79±1.97	**44.46±2.41**
	OA	54.61±2.90	60.62±1.59	67.27±1.81	74.42±1.46	74.34±0.70	73.45±1.87	81.94±2.63	69.12±0.87	69.35±1.02	**84.41±0.86**
	AA	73.07±1.10	78.84±1.01	83.30±0.98	87.17±0.63	87.90±0.51	87.07±0.83	91.12±1.38	81.60±0.58	84.31±0.39	**92.31±0.83**
	Kappa	49.02±2.91	55.55±1.64	63.12±1.84	73.68±1.45	70.97±0.72	69.92±1.95	79.28±3.01	64.93±0.92	65.13±1.06	**81.97±1.01**

weight parameters. On different datasets, we randomly select several classes as novel classes, and the remaining classes as known classes. Specifically, we set classes 3 and 6 on the IP dataset, classes 4, 12, and 14 on the SA dataset, and class 9 on the PU dataset as novel classes. We randomly select 10 and 20 hyperspectral pixels in each known class as labeled pixels on three datasets. On the IP dataset, due to the limited number of hyperspectral pixels in the class "grass-pasture-mowed" and class "oats" (class 7 and class 9), we randomly selected 10 pixels in these known classes as labeled pixels in all experiments. The experiments, including the proposed method and the state-of-the-art compared methods, are performed ten times, and the average values of experimental results are treated as their classification performance. In addition, the proposed CACL is implemented using the PyTorch framework with version 1.11, and different methods, including our proposed CACL and other methods, are trained on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU with 24-GB memory.

C. Experimental Result

1) IP Dataset: We conduct extensive experiments on the IP dataset to verify the effectiveness of our CACL, and the results are listed in Table I. SpectralFormer and SSFTT employ transformer-based backbone networks to extract features from labeled samples, and do not consider unlabeled samples containing novel classes during the model training process, limiting feature discrimination when labeled samples are small. The accuracy of novel classes with ten labeled samples in each known class is 11.09% and 6.19%. MDL4OW and SSMLP-RPL do not utilize the spectral–spatial information of unlabeled samples to optimize their model's weight parameters, which limits classification accuracy. The OAs of these methods with ten labeled samples in each known class are 57.53% and 66.01%. Due to FullyContNet, AMGCFN, and SGMAEs extracting spectral–spatial features from the entire HSI using CNNs or GNNs, unlabeled samples belonging to novel classes may confuse the model's discrimination, reducing the classification performance. The OAs of these methods with ten labeled samples in each known class are 66.84%, 75.06%, and 59.95%, respectively. Our proposed CACL develops a known class sampler to accurately select unlabeled samples belonging to known classes, and then effectively exploits the spectral and spatial information of these samples by constraining different views to produce consistent predictions. As shown in Table I, our proposed CACL achieves higher classification accuracy in both known and novel classes. On the accuracy of novel classes, OA, AA, and Kappa, the performance of the proposed method with 10 and 20 labeled samples in each known class is (65.67%, 81.65%, 88.92%, and 78.78%) and (44.46%, 84.41%, 92.31%, and 81.97%), respectively. In classes 2, 5, 9, 10, 11, and 12, the proposed method achieves the best classification accuracy with ten labeled samples in each known class. As the number of labeled samples increases, i.e., 20 labeled samples in each

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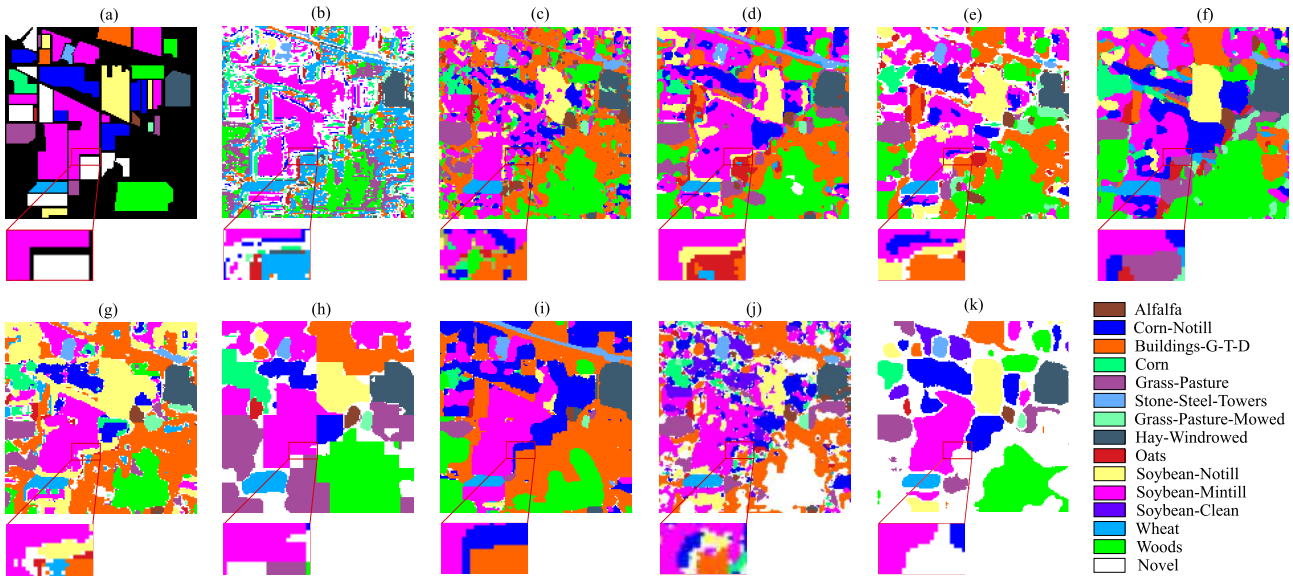


Fig. 4. Classification maps of different methods on the IP dataset. (a) Ground truth. (b) SpectralFormer. (c) SSFTT. (d) MDL4OW. (e) SSMLP-RPL. (f) FullyContNet. (g) PUSL. (h) AMGCFN. (i) RSEN. (j) SGMAEs. (k) CACL.

known class, different methods can more effectively exploit the spectral–spatial features of the HSI, and their classification performance is significantly improved. In addition, compared with other methods, CACL's standard deviation in OA, AA, and Kappa also achieves competitive results, indicating that our method has strong stability.

Fig. 4 shows the classification maps of different methods under ten labeled samples in each known class. Different land-cover classes within the IP dataset have complex boundary information, which poses great challenges to extracting discriminative features when the number of labeled samples is small. SpectralFormer applies the transformer to extract relations among spectral bands. It fails to extract discriminative features under a few labeled samples. FullyContNet, AMGCFN, and SGMAEs extract features from the entire HSI. They are easily affected by background pixels whose classes are inconsistent with their center pixels, thereby reducing classification accuracy in both known and novel classes. As shown in Fig. 4, it can be observed that our method accurately identifies known and novel classes in the zoomed-in view area, and the classification map is smoother compared to other methods.

2) SA Dataset: Similar to the IP dataset, we conduct extensive experiments on the SA dataset to verify the effectiveness of our proposed CACL, and the results are illustrated in Table II. MDL4OW only utilizes labeled samples to train the weight parameters of its encoder–decoder network, and introduces the reconstruction errors of unlabeled samples to reject novel classes. The classification accuracy is limited under a small number of labeled samples, and the OA, AA, and Kappa with ten labeled samples in each known class are 73.36%, 80.26%, and 70.43%, respectively. PUSL generates pseudo-labels for unlabeled samples based on confidence scores, and these pseudo-labels are used to optimize the model's parameters. However, PUSL easily treats unlabeled samples from novel classes as known classes when a few labeled samples are available. The performance on OA and novel classes with ten labeled samples in each known class is 83.41% and 4.19%, respectively. The proposed CACL employs consistency learning to efficiently leverage the spectral–spatial information of unlabeled samples from known classes, and introduces a PCO to improve interclass distinguishability and intraclass compactness, thereby achieving the best classification performance. On the accuracy of novel classes, OA, AA, and Kappa, the performance of the proposed method with 10 and 20 labeled samples in each known class is (41.40%, 88.61%, 91.27%, and 87.33%) and (40.88%, 90.60%, 93.16%, and 89.53%), respectively. When there are 20 labeled samples in each known class, CACL also achieves the best performance in class 8. In terms of standard deviation with 20 labeled samples per known class, our CACL achieves competitive results with most compared methods, and outperforms SpectralFormer, SSFTT, and MDL4OW.

Fig. 5 shows the classification maps of different methods under ten labeled samples in each known class. Land-cover classes have similar spectral bands (e.g., class "grapes_untrained" and class "vinyard_untrained") and complex spatial position distribution information (e.g., zoomed-in view areas) on the SA dataset. SpectralFormer, SSFTT, MDL4OW, and SSMLP-RPL fail to learn discriminative features under limited labeled samples. FullyContNet, PUSL, AMGCFN, RSEN, and SGMAEs are easily disturbed by unlabeled samples from novel classes during the model training process, thereby reducing feature distinguishability and limiting classification accuracy. The proposed CACL effectively models spatial position priors between pixels within HSI cubes via the proposed symmetric PAE, and introduces the PCO to improve the distinguishability between known classes by decreasing the distance of samples from the same class and expanding those from different classes in the feature domain, achieving the best classification accuracy. As shown in Fig. 5, the proposed method can well predict class

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TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE SA DATASET

Size	Class	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	FullyContNet	PUSL	AMGCFN	RSEN	SGMAEs	CACL
10	1	91.85	99.13	95.28	97.83	94.97	96.85	**100.00**	99.02	99.72	97.50
	2	95.76	98.53	75.73	99.87	97.77	98.64	99.87	99.32	**99.99**	99.88
	3	64.39	90.39	83.16	94.52	**100.00**	97.82	**100.00**	97.72	99.91	97.09
	5	92.05	98.37	92.67	97.85	97.83	97.84	95.44	**99.88**	97.02	93.65
	6	99.52	99.81	83.56	98.58	98.59	**100.00**	99.19	99.97	99.87	95.40
	7	87.96	99.84	98.88	99.51	97.79	97.68	99.86	98.96	**99.95**	98.41
	8	45.22	74.61	60.83	68.13	68.73	80.16	**89.60**	7.94	54.98	85.62
	9	90.99	**100.00**	96.63	97.58	99.90	98.56	**100.00**	99.63	99.78	97.40
	10	61.62	93.36	84.12	89.52	**98.29**	88.02	96.92	96.95	95.20	92.80
	11	56.87	98.00	91.66	95.44	98.97	96.84	**100.00**	99.36	99.75	99.26
	13	99.54	**100.00**	**100.00**	99.96	98.36	**100.00**	99.89	**100.00**	**100.00**	97.88
	15	66.88	61.80	63.97	82.22	94.80	77.16	64.13	**98.36**	85.96	87.51
	16	77.83	97.50	93.71	97.38	**100.00**	96.59	**100.00**	99.92	99.25	94.01
	Novel	10.11±4.94	4.94±2.49	3.52±6.22	34.42±15.66	0.00±0.00	4.19±3.85	10.04±7.76	0.00±0.00	5.89±2.08	**41.40±3.21**
	OA	68.78±5.05	80.76±0.88	73.36±2.95	84.04±2.02	83.82±1.50	83.41±1.81	85.23±1.71	72.01±0.77	80.59±0.50	**88.61±1.76**
	AA	74.33±3.08	86.88±1.56	80.26±2.11	89.49±1.03	89.00±0.66	87.88±1.14	89.64±1.17	85.50±0.36	88.38±0.20	**91.27±1.48**
	Kappa	65.52±5.41	78.56±0.98	70.43±3.24	82.26±2.17	82.07±1.64	81.52±2.01	83.52±1.93	69.32±0.83	78.49±0.55	**87.33±1.95**
20	1	93.28	99.87	97.09	99.37	**99.90**	99.10	99.80	99.43	99.74	98.76
	2	99.60	99.62	96.91	99.87	94.90	99.48	**100.00**	99.98	99.99	99.33
	3	81.48	99.85	94.23	98.15	98.53	96.69	**100.00**	99.51	99.93	98.48
	5	95.27	98.45	93.07	**99.64**	98.81	94.82	89.99	99.39	99.16	98.78
	6	99.91	**99.98**	85.34	98.48	99.97	99.94	99.39	99.92	99.93	98.67
	7	91.60	99.83	97.10	99.69	99.53	98.12	93.41	**100.00**	99.96	98.09
	8	69.13	75.23	70.15	79.11	77.42	85.67	86.13	85.38	79.14	**87.23**
	9	95.39	99.91	97.92	99.36	99.81	99.73	99.29	**99.98**	99.96	98.82
	10	86.83	95.22	90.34	92.66	98.32	92.92	**99.15**	98.68	95.46	96.52
	11	91.18	99.55	95.74	99.47	**100.00**	98.86	94.01	99.52	99.98	99.47
	13	99.72	**100.00**	**100.00**	99.96	98.03	99.98	**100.00**	**100.00**	**100.00**	99.11
	15	52.97	83.92	75.55	84.96	**93.00**	84.16	80.43	89.40	89.06	92.28
	16	87.52	98.09	97.25	98.32	97.51	96.89	**100.00**	**100.00**	99.21	97.78
	Novel	6.96±2.94	1.77±0.70	19.54±5.58	33.78±6.97	0.00±0.00	2.56±3.65	26.67±5.13	0.00±0.00	5.75±1.26	**40.88±1.93**
	OA	76.04±1.58	84.22±2.67	80.84±1.99	87.44±0.56	85.50±1.39	85.82±0.78	87.30±1.15	87.25±1.04	86.17±0.46	**90.60±1.50**
	AA	82.20±0.90	89.38±0.81	86.44±1.45	91.63±0.46	89.69±0.68	89.21±0.59	90.59±0.53	90.80±0.33	90.52±0.17	**93.16±1.25**
	Kappa	73.33±1.73	82.46±2.89	78.71±2.19	86.01±0.63	83.91±1.54	84.17±0.85	85.87±1.27	85.83±1.13	84.64±0.50	**89.53±1.67**

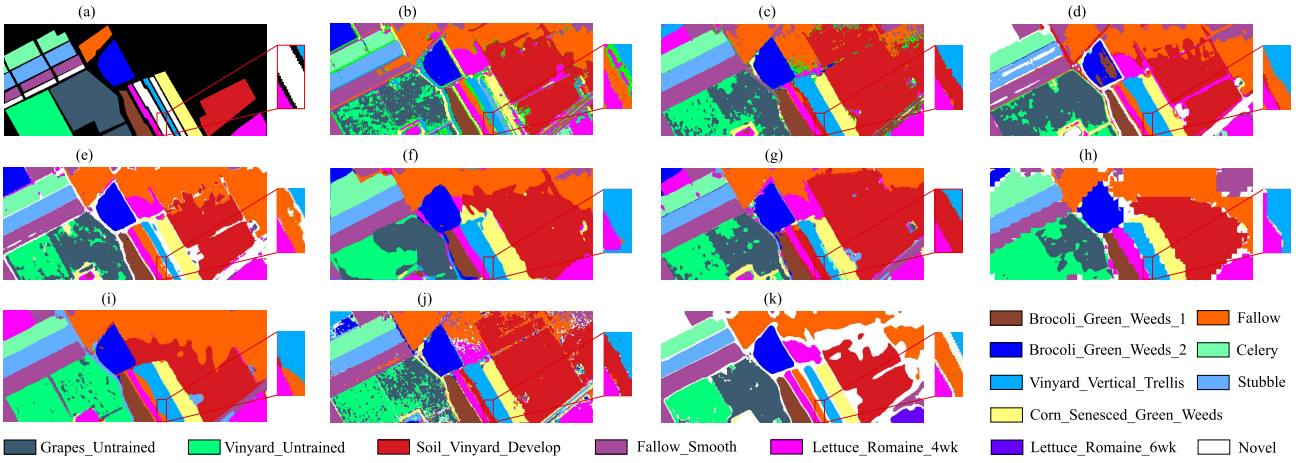


Fig. 5. Classification maps of different methods on the SA dataset. (a) Ground truth. (b) SpectralFormer. (c) SSFTT. (d) MDL4OW. (e) SSMLP-RPL. (f) FullyContNet. (g) PUSL. (h) AMGCFN. (i) RSEN. (j) SGMAEs. (k) CACL.

"Lettuce_Romaine_4wk" and novel classes in the zoomed-in view area.

3) PU Dataset: Several experiments on the PU dataset are conducted to analyze the performance of different methods, and the results are summarized in Table III. Compared to the IP and SA datasets, hyperspectral pixels in the PU dataset show more complex spatial position distribution information. FullyContNet, AMGCFN, and SGMAEs, extracted features from the entire HSI, are easily perturbed by novel class samples. The performance of these methods on novel classes is 0.15%, 9.12%, and 4.87%, respectively. Since PUSL is prone to overfitting under limited labeled samples, it easily treats unlabeled samples from novel classes as reliable and unreliable samples for training, which confuses the model discrimination. The accuracy of novel classes with 10 and 20 samples in each known class is 2.29% and 1.84%, respectively. Our proposed CACL can effectively extract robust and discriminative spectral–spatial features by modeling the spatial position priors between the center pixel and its neighboring pixels, and fully utilizes unlabeled samples belonging to known classes generated from the proposed known class sampler, achieving the best classification results. On the accuracy of novel classes, OA, AA, and Kappa, the performance of the proposed method with 10 and 20 labeled samples in each

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TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE PU DATASET

Size	Class	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	FullyContNet	PUSL	AMGCFN	RSEN	SGMAEs	CACL
10	1	50.11	76.16	79.49	85.89	84.24	80.39	88.68	81.67	82.22	**93.42**
	2	77.11	80.11	77.56	78.52	74.22	81.62	85.58	66.96	**96.10**	92.55
	3	54.41	86.32	71.79	82.76	75.60	71.94	**94.07**	77.84	75.14	81.60
	4	84.18	92.02	92.43	93.47	95.35	91.73	89.22	**96.42**	87.41	93.05
	5	97.53	99.99	75.73	99.91	99.81	99.99	**100.00**	99.88	99.91	99.59
	6	48.70	82.51	76.80	82.20	89.19	73.61	**96.62**	72.66	65.94	96.56
	7	79.28	98.80	86.72	95.31	93.10	82.43	98.42	90.56	**99.77**	97.61
	8	64.95	57.03	67.81	84.50	89.76	84.44	**97.51**	75.57	63.00	96.70
	Novel	12.16±15.70	0.72±0.82	2.11±3.46	0.69±1.04	0.15±0.21	2.29±2.18	9.12±3.52	0.00±0.00	4.87±3.23	**69.67±4.99**
	OA	67.20±3.91	78.40±3.60	76.27±5.29	83.41±4.38	80.21±1.76	79.83±2.84	88.22±4.17	73.58±5.42	84.12±1.04	**92.88±1.83**
	AA	63.16±2.01	74.85±1.75	70.05±2.49	78.14±1.89	77.94±1.62	74.27±2.06	84.36±1.83	73.51±1.69	74.93±0.87	**91.20±1.59**
	Kappa	57.70±4.11	72.31±4.22	69.66±6.29	76.19±5.05	74.89±2.11	74.05±3.68	84.91±5.01	66.82±5.92	78.77±1.32	**90.78±2.32**
20	1	64.21	85.59	85.99	93.31	89.90	89.07	**96.36**	92.24	92.72	96.19
	2	83.69	87.29	86.56	91.43	91.75	93.62	96.25	85.12	94.11	**97.08**
	3	46.06	83.77	81.19	92.58	90.00	85.68	**98.59**	88.04	75.44	97.59
	4	89.39	95.29	93.41	95.81	**96.85**	94.46	94.41	96.56	90.89	95.01
	5	98.91	**100.00**	78.68	99.80	**100.00**	**100.00**	**100.00**	**100.00**	99.96	99.90
	6	47.26	87.62	87.17	94.28	92.16	86.63	**99.92**	85.11	83.59	99.27
	7	76.13	98.62	93.52	97.66	98.09	93.11	99.91	93.08	98.17	**100.00**
	8	78.66	74.33	84.15	88.39	94.63	90.30	**98.41**	94.98	76.12	97.56
	Novel	15.14±26.47	0.13±0.15	1.05±0.71	0.31±0.31	0.021±0.07	1.84±1.28	10.07±6.02	0.00±0.00	0.05±0.11	**64.24±3.64**
	OA	73.24±2.24	85.17±2.81	84.64±2.45	93.37±2.16	90.46±3.00	89.63±1.74	95.19±2.34	86.87±3.01	88.19±0.90	**96.57±1.03**
	AA	66.61±3.67	79.18±1.65	76.86±2.20	83.73±0.86	83.71±1.70	81.63±1.00	88.21±0.99	81.68±0.61	79.00±0.71	**94.09±1.51**
	Kappa	64.89±2.50	80.73±3.47	80.01±3.14	87.79±2.72	87.54±3.75	86.32±2.28	93.68±2.95	82.96±3.66	84.35±1.16	**95.50±1.34**

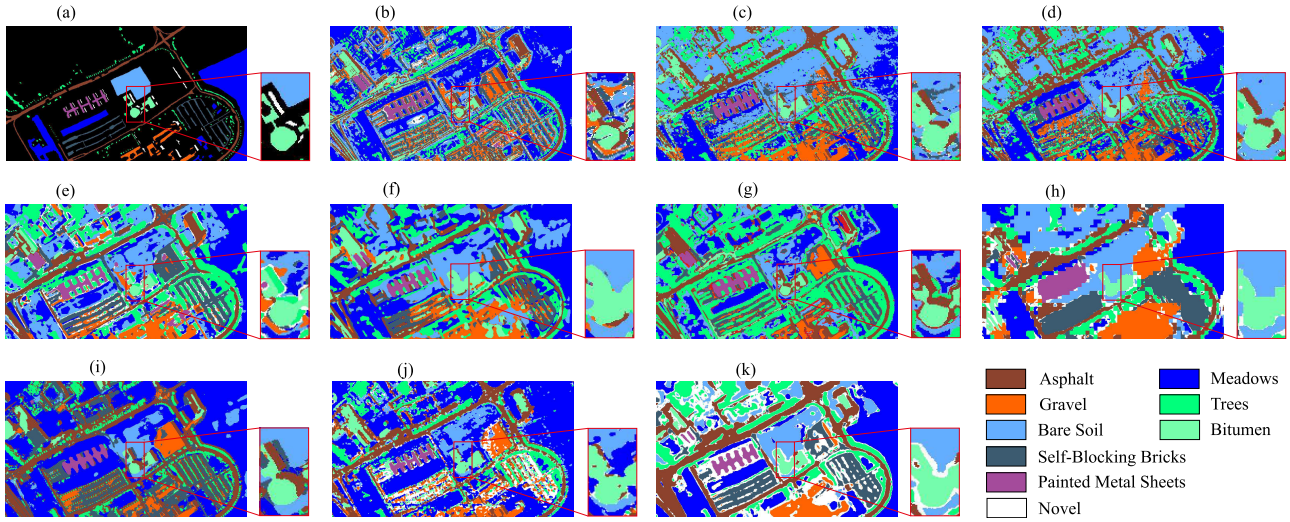


Fig. 6. Classification maps of different methods on the PU dataset. (a) Ground truth. (b) SpectralFormer. (c) SSFTT. (d) MDL4OW. (e) SSMLP-RPL. (f) FullyContNet. (g) PUSL. (h) AMGCFN. (i) RSEN. (j) SGMAEs. (k) CACL.

known class is (69.67%, 92.88%, 91.20%, and 90.78%) and (64.24%, 96.57%, 94.09%, and 95.50%), respectively. With a small number of labeled samples (ten labeled samples in each known class), the classification result of the proposed method in class 1 significantly outperforms the compared methods. In addition, similar to the IP and SA datasets, the proposed method also shows good stability and strong practicality.

Fig. 6 shows the classification maps of different methods under ten labeled samples in each known class. On the one hand, as complex spatial position distribution information and spectral bands of the HSI, most compared methods struggle to extract discriminative features when the number of labeled samples is insufficient. On the other hand, these methods overlook that unlabeled samples contain novel classes during the training and testing stages. Similar to the IP and SA datasets, we observe that the proposed CACL achieves higher classification accuracy than other methods in the zoomed-in view area, and accurately classifies both known and novel classes.

D. Ablation Study

1) Impact of Proposed PAT: Existing transformer-based methods often model spatial position information between input tokens within HSI cubes via a learnable PE. However, these methods fail to utilize the spatial position priors, resulting in the underutilization of HSI information. Our CACL can effectively model the spatial position priors between pixels using the proposed symmetric PAE. We perform extensive experiments to prove the effectiveness of our proposed PAT on performance, and the results with ten labeled samples in each known class are illustrated in Table IV. We observe that the classification performance of CACL drops sharply when

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TABLE IV
IMPACT OF THE PROPOSED PAT ON CLASSIFICATION PERFORMANCE (%)

Dataset	PE	PAE	Novel	OA	AA	Kappa
IP	✔		55.06	78.77	87.00	75.52
		✔	65.67	81.65	88.92	78.78
SA	✔		23.56	86.16	89.80	84.63
		✔	41.40	88.61	91.27	87.33
PU	✔		47.18	91.97	87.78	89.55
		✔	69.67	92.88	91.20	90.78

TABLE V
IMPACT OF THE PROPOSED PDCL ON CLASSIFICATION PERFORMANCE (%)

Dataset	Method	Novel	OA	AA	Kappa
IP	CACL w/o PDCL	56.04	76.54	87.02	73.12
	CACL w/ PDCL	35.15	78.55	88.54	75.28
SA	CACL w/o PDCL	28.94	87.01	90.58	85.56
	CACL w/ PDCL	23.18	87.72	90.57	86.33
PU	CACL w/o PDCL	51.89	84.62	85.78	80.49
	CACL w/ PDCL	46.42	91.97	90.17	89.70

TABLE VI
IMPACT OF THE PROPOSED PCO ON CLASSIFICATION PERFORMANCE (%)

Dataset	Method	Novel	OA	AA	Kappa
IP	CACL w/o PCO	35.15	78.55	88.54	75.28
	CACL	65.67	81.65	88.92	78.78
SA	CACL w/o PCO	23.18	87.72	90.57	86.33
	CACL	41.40	88.61	91.27	87.33
PU	CACL w/o PCO	46.42	91.97	90.17	89.70
	CACL	69.67	92.88	91.20	90.78

our proposed PAE is replaced by PE. In terms of OA, our proposed CACL decreased by 2.88%, 2.45%, and 0.91% on the IP, SA, and PU datasets, respectively.

2) Impact of Proposed PDCL: To avoid novel class samples interfering with the known class discrimination, the PDCL is developed to accurately select unlabeled samples belonging to known classes using the proposed known class sampler, and then fully explore their spectral–spatial information by producing consistent predictions across different views. We conduct extensive experiments to explore the impact of our proposed PDCL. The experimental results with ten labeled samples in each known class are listed in Table V, where CACL w/o PDCL represents that uses only labeled samples to optimize the CACL's parameters, and CACL w/PDCL denotes that exploits the spectral–spatial information of unlabeled samples from known classes. We observe that selectively utilizing the spectral–spatial information of unlabeled samples can improve classification accuracy. Compared with CACL w/o PDCL, in terms of OAs, CACL w/PDCL improves by 2.01%, 0.71%, and 7.35% on three datasets, respectively.

3) Impact of Proposed PCO: Samples from the same class are closer in the feature domain, and those from different classes are separated. To accurately predict known and novel classes, the proposed CACL utilizes labeled sample features to construct class prototypes for different known classes and employs the PCO to further improve interclass distinguishability and intra-class compactness. We conduct extensive experiments to analyze the impact of the proposed PCO on performance. The experimental results with ten labeled samples in each known class are listed in Table VI, where CACL w/o PCO means removing PCO in CACL. As shown in Table VI, compared with CACL w/o PCO, our CACL achieves higher performance in known and novel classes. In terms of OA, the proposed CACL improves by 3.10%, 0.89%, and 0.91%, on the IP, SA, and PU datasets, respectively.

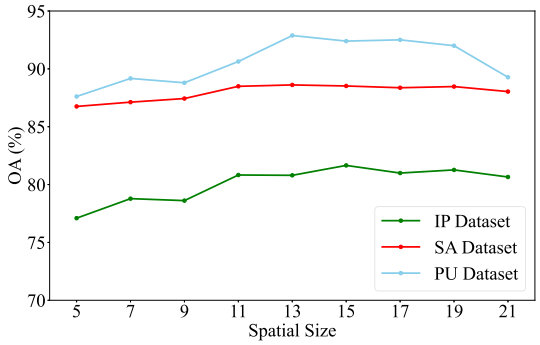


Fig. 7. Classification performance of CACL with different spatial sizes.

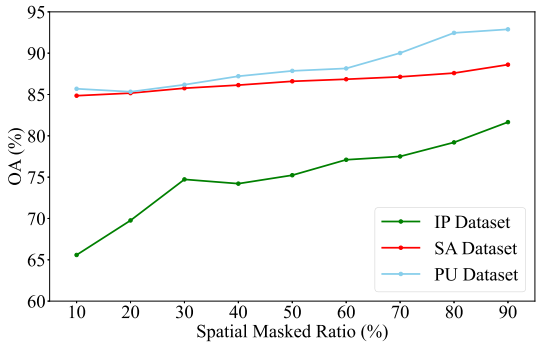


Fig. 8. Classification performance of CACL with different spatial masked ratios.

E. Parameter Analysis

1) Impact of Spatial Size: A larger HSI cube will contain more neighboring pixels, which can provide richer spectral–spatial information to classify the land-cover classes of the center pixels. However, a too-large spatial size may lead to several neighboring pixels belonging to other classes, which will confuse feature discrimination and reduce classification accuracy. We conduct extensive experiments to explore the impact of the spatial size on performance, and the results with ten labeled samples in each known class are listed in Fig. 7. On the IP, SA, and PU datasets, the proposed CACL achieves the best classification accuracy when the spatial size is set to 15×15, 11×11, and 13×13, respectively.

2) Impact of Spatial Masked Ratio: The proposed method masks some pixel tokens in the spatial dimension as the strong augmentation view, and regards the original pixel tokens as the weak augmentation view. We perform extensive experiments to

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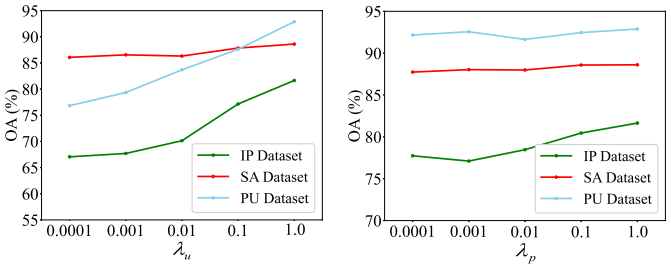


Fig. 9. Classification performance (%) of CACL with different tradeoff parameters.

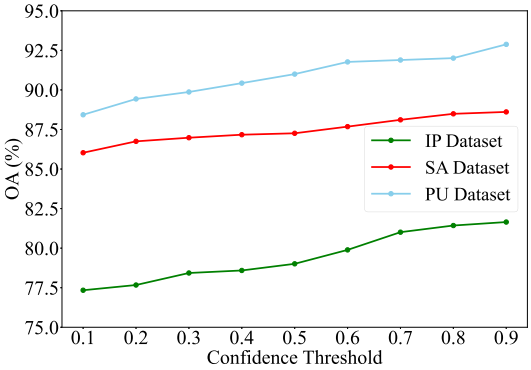


Fig. 10. Classification performance of CACL with different confidence thresholds.

explore the impact of the spatial masked ratio on performance, and the results with ten labeled samples in each known class are illustrated in Fig. 8. A smaller spatial masked ratio may cause HSI cubes to contain more redundant information, while a larger spatial masked ratio will lose too much information, which limits feature discrimination. We observe that our CACL achieves the optimal classification accuracy, when the spatial masked ratio is set to 90% on three datasets.

3) Impact of Tradeoff Parameters λ_u and \lambda$_p$: The proposed CACL combines supervised loss, class-aware consistency loss, and PCO loss to optimize the model's parameters. To analyze the impact of tradeoff parameters λ_u and λ_p on performance, we conduct extensive experiments on IP, SA, and PU datasets, and the results with ten labeled samples in each known class are listed in Fig. 9. On three datasets, when λ_u and λ_p are set to 1, the proposed CACL achieves the best classification accuracy.

4) Impact of Confidence Threshold \gamma: A larger confidence threshold γ means that more unlabeled samples belonging to known classes will be discarded, leading to insufficient utilization of spectral–spatial information. Conversely, a smaller confidence threshold γ may introduce more unlabeled samples from novel classes, thereby confusing model discrimination and reducing feature distinguishability between known classes. To explore the impact of the confidence threshold γ on classification performance, we conduct a large number of experiments on the IP, SA, and PU datasets, and the experimental results with ten labeled samples in each known class are illustrated in Fig. 10. It can be observed that the proposed method achieves the best performance when the confidence threshold γ is set to 0.9 on the IP, SA, and PU datasets.

TABLE VII

CLASSIFICATION PERFORMANCE (%) OF CACL WITH DIFFERENT LABELED SAMPLE SIZES

Dataset	Size	5	10	15	20	30	50
IP	OA	72.64	81.65	82.93	84.41	85.20	87.83
	AA	80.65	88.92	90.57	92.31	92.28	92.69
	Kappa	68.31	78.78	80.27	81.97	82.89	85.90
SA	OA	84.00	88.61	90.05	90.60	92.01	94.27
	AA	86.52	91.27	92.47	93.16	93.97	95.17
	Kappa	82.20	87.33	88.91	89.53	91.09	93.60
PU	OA	78.99	92.88	94.64	96.57	97.66	98.34
	AA	81.73	91.20	91.62	94.09	93.04	93.28
	Kappa	73.78	90.78	93.05	95.50	96.89	97.78

TABLE VIII

IMPACT OF PIXELS FROM NOVEL CLASSES ON CLASSIFICATION PERFORMANCE (%)

Dataset	Method	Novel	OA	AA	Kappa
IP	CACL w/ n_pixels	37.88	78.78	89.04	75.63
	CACL	65.67	81.65	88.92	78.78
SA	CACL w/ n_pixels	23.31	87.82	90.92	86.46
	CACL	41.40	88.61	91.27	87.33
PU	CACL w/ n_pixels	23.44	89.76	86.88	86.91
	CACL	69.67	92.88	91.20	90.78

5) Impact of the Number of Labeled Samples: The proposed method constructs class prototypes for each category using the spectral–spatial features of labeled samples to accurately select unlabeled samples belonging to known classes. These retained unlabeled samples and a small number of labeled samples are further used to optimize the model's weight parameters. To explore the performance of the proposed method with different numbers of labeled samples, we perform extensive experiments on the IP, SA, and PU datasets, including {5, 10, 15, 20, 30, 50} labeled samples in each known class. The experimental results are shown in Table VII. When there are only a small number of labeled samples, the proposed method struggles to effectively model the complex data distribution of the HSI, thus limiting classification performance. As the number of labeled samples increases, the proposed method not only constructs discriminative class prototypes to filter novel class samples but also provides richer supervised information for model optimization. As shown in Table VII, as the number of labeled samples increases, the proposed method achieves higher classification performance.

6) Impact of Unlabeled Samples From Novel Classes: To reduce the risk of novel class samples interfering with feature discrimination, the proposed method selectively utilizes the spectral–spatial information of unlabeled samples. To validate the effectiveness of the proposed method, we conduct extensive experiments on the IP, SA, and PU datasets. The results are illustrated in Table VIII, where CACL w/ n_pixels represents that all unlabeled samples, including those from both known and novel classes, are used to optimize the model's weight parameters. According to the results in Table VIII, we

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TABLE IX
CLASSIFICATION PERFORMANCE (%) OF CACL UNDER DIFFERENT NOVEL CLASSES

Novel Classes	Openness	Size	10	20
3, 6	3.39	Novel	65.67	44.46
		OA	81.65	84.41
		AA	88.92	92.31
		Kappa	78.78	81.97
5, 6, 12	5.31	Novel	51.25	44.87
		OA	77.05	83.29
		AA	87.78	91.83
		Kappa	73.36	80.48
4, 8, 10, 13, 15	9.73	Novel	37.80	33.99
		OA	74.39	79.66
		AA	85.26	90.74
		Kappa	69.86	76.22

TABLE X
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON THE IP DATASET

Method	Training Time (s)	Testing Time (s)	Parameters (M)
SpectralFormer	34.41	0.81	0.362
SSFTT	12.15	0.44	0.153
MDL4OW	34.50	0.87	0.420
SSMLP-RPL	347.31	18.42	10.460
FullyContNet	488.46	0.11	1.255
PUSL	905.58	0.73	0.250
AMGCFN	18.56	0.21	0.368
RSEN	64.53	0.30	1.337
SGMAEs	2.95	0.13	0.120
CACL	948.83	3.80	0.113

observe that the proposed method achieves higher accuracy compared to CACL w/ n_pixels. On the IP, SA, and PU datasets, the OAs of our CACL improve 2.87%, 0.79%, and 3.12%, respectively. Moreover, since CACL w/ n_pixels utilizes unlabeled samples belonging to novel classes for model optimization, it confuses the distinguishability between known classes, and the accuracy of novel classes also drops sharply.

7) Impact of Openness: As a common evaluation criterion, openness [47], [62] describes the ratio between known and novel classes, and it can be used to measure the model's generalization and robustness. A larger openness value indicates a larger number of novel classes in unlabeled samples. To explore the performance of the proposed CACL under different openness, we conduct extensive experiments on the IP dataset, and the experimental results and settings are listed in Table IX. Our CACL effectively selects known class samples using the proposed known class sampler, reducing the interference from novel class samples. As shown in Table IX, it can be observed that the proposed CACL achieves promising performance in both known and novel classes under different openness.

F. Complexity Analysis

To analyze the computational complexities of different methods, including training time, inference time, and the number of parameters, we conduct a large number of experiments on the IP dataset, and the results of different methods are illustrated in Table X. SpectralFormer and SSFTT optimize the model's weight parameters by only using a small number of labeled samples. They consume less training time. Compared to SpectralFormer and SSFTT, since MDL4OW incorporates the reconstruction task and SSMLP-RPL introduces the reciprocal points learning, these methods require higher training and testing times. AMGCFN divides the HSI into a series of superpixels for feature extraction, which reduces the training cost compared to the proposed CACL. The remaining methods, including FullyContNet, PUSL, SGMAEs, and our CACL, require higher training time as they utilize both labeled and unlabeled samples simultaneously. Overall, the proposed CACL achieves excellent classification performance while maintaining competitive computational efficiency.

V. CONCLUSION

This article proposes a CACL method for open-set semi-supervised HSI classification, which selectively utilizes the spectral–spatial information of unlabeled samples to optimize the model. First, to extract robust and discriminative features, a PAT is developed, which effectively models spatial position priors between the center pixel and its neighboring pixels through the proposed symmetric PAE. Second, to avoid novel class samples interfering with the distinguishability among known classes, a PDCL is introduced, which accurately selects unlabeled samples belonging to known classes through the proposed known class sampler, and effectively leverages their spectral–spatial information by producing consistent predictions across different views. Finally, a PCO is proposed to improve interclass distinguishability and intra-class compactness by constraining the distance of hyperspectral data in the feature domain. Moreover, an adaptive segmentation threshold is proposed to predict known classes and reject novel ones. Extensive experiments on the IP, SA, and PU datasets have demonstrated that the proposed CACL achieves superior performance in both known and novel classes, and significantly surpasses other state-of-the-art HSI classification methods.

In the future, we plan to investigate semi-supervised classification methods that can adaptively handle hyperspectral data acquired from various remote sensing scenes, improving the model's generalization and practicality. To accurately predict the land-cover classes of hyperspectral pixels, existing methods typically train classification models for specific remote sensing scene data. These models fail to directly transfer to other remote sensing scenes, which limits their generalization and practicality in real-world applications. To address this issue, a self-supervised learning method is a promising solution. Specifically, unlabeled samples from various remote sensing scenes can be used to pretrain the model's weight parameters, enabling this model to learn general spectral–spatial features. To further extract the specific characteristics, a small number of labeled samples are then utilized to finetune, thereby improving classification accuracy.

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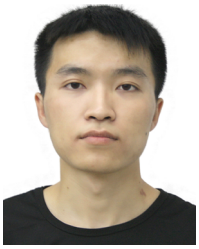
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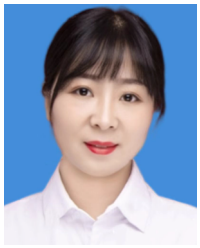
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